



Linux Storage Practice Notes

1 Purpose of `/dev` Directory

- `/dev` is a special directory in Linux.
- It contains **device files**.
- These files represent hardware like disks, CPU, memory, etc.

Examples:

- `/dev/nvme0n1` → Main disk (root disk)
- `/dev/nvme1n1` → Additional disk (new volume)

👉 These are **not normal files**.

👉 They are used by the system to **interact with hardware**.

2 Disk Formatting and Mounting

Before using a new disk, we must follow these steps:

Step 1: Format the Disk

```
mkfs.ext4 /dev/nvme1n1
```

- Creates a filesystem (ext4)
 - Makes the disk ready to store data
-

Step 2: Mount the Disk

```
mount /dev/nvme1n1 /mnt
```

- Attaches the disk to a folder
 - Now you can access disk data using that folder
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3 Why We Use `/mnt` (Temporary Mount Point)

- `/mnt` is used for **temporary mounting**
- It is safe for testing new disks
- Helps prevent data loss

👉 If you mount directly to important folders (like `/var`, `/usr`):

- Old data can become hidden
 - The system may break
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4 Best Practice (DevOps Way)

- Mount new disk to `/mnt`
 - Verify the disk is working
 - Copy data (if needed)
 - Then mount to the final location
-

✅ Final Summary

- `/dev` → Contains device files
- `mkfs.ext4` → Formats disk
- `mount` → Attaches disk to folder
- `/mnt` → Safe temporary mount location

👉 Always test before final mounting.
